

Duncan Park

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EDUCATION

Cornell University, College of Arts and Sciences

B.A. in Mathematics and Computer Science

Courses: Data Structures and Functional Programming, Real Analysis, Mathematical Foundations of CS

Stuyvesant High School • *Diploma received June 2025*

Ithaca, NY

Expected May 2029

GPA: 3.67/4.0

New York, NY

TECHNICAL SKILLS

Languages: Python, Java, OCaml, R

ML & Data Science: PyTorch, scikit-learn, pandas, NumPy, SciPy, tidyverse, ggplot2, Autoencoders, Representation Learning

Bioinformatics & Stats: Mixed-Effect Models, Survival Analysis, ANOVA, MAFFT, IQ-TREE, BEAST2

Dev Tools: Git, GitHub, VS Code, IntelliJ, RStudio, Jupyter Notebook

RELEVANT EXPERIENCE

MA'AYAN LAB - COMPUTATIONAL SYSTEMS BIOLOGY AT MOUNT SINAI

New York, NY

Summer Trainee

6/2026 – 8/2026

- Developed a PyTorch multimodal encoder integrating chemical fingerprints and LINCS gene-expression signatures to predict 83 adverse drug reactions (ADRs); outperformed a chemical-only baseline.
- Engineered preprocessing and representation pipelines for 16M+ FAERS patient reports; standardized 20K+ drug names via PubChem and encoded multi-drug combinations of up to five drugs.
- Applied representation learning and neural networks using Python, PyTorch, NumPy, and SciPy; presented methodology, model results, and biological findings to Mount Sinai faculty at the 2026 Biomedical Big Data Science Symposium.

BIG RED SPORTS NETWORK - FOOTBALL ANALYTICS

Ithaca, NY

Analyst, Research and Development Sub-Team Member

8/2025 – Present

- Produced 10 weekly scouting reports for Cornell football by analyzing tendencies and visualizing data in heat maps/tables.
- Collaborated with R&D team to build tempo and punting performance models from scratch to track performance metrics.

CORNELL SPORTS ANALYTICS CLUB

Ithaca, NY

Social Chair, Analyst

8/2025 – Present

- Built and refined ML model predicting field-goal outcomes in the NFL and presented findings to 30 members and leaders.
- Organized watch parties and dinners while managing social media to promote members' research and showcase projects.

PROJECT EXPERIENCE

ANALYZING IMPACT OF WORKLOAD ON INJURY RISK IN THE NFL

Ithaca, NY

STSCI 4981 - Sports Analytics in Practice, Cornell University

3/2026 – 5/2026

- Used nflreadpy to extract NFL player and schedule data to engineer workload and efficiency features for skill positions.
- Built linear mixed-effect models and survival curves in R to assess how efficiency metrics evolved across workload groups.
- Conducted ANOVA and Chi-squared tests in R that showed high touch volume does not signify short-term injury risk.

MARKOV RANKINGS ACROSS SPORTS LEAGUES

New York, NY

Linear Algebra, Stuyvesant High School

4/2025 – 6/2025

- Built stochastic matrices using PageRank concepts in R to model steady-state team ratings in the NFL, NHL, and NBA.
- Engineered Monte Carlo simulation engine with Softmax probabilities and home-field adjustments to forecast outcomes.
- Evaluated predictive accuracy against actual results, achieving Normalized Mean Rank Error of 0.0968 for NHL and NFL.

CORNELL DATA SCIENCE PROJECT TEAM

Ithaca, NY

Machine Learning Engineer

9/2026 – Present

- Selected as 1 of 3 Machine Learning Engineers for Fall 2026 cohort; participating in machine/deep learning training.

LEADERSHIP

NYC TEENS FOR 9 MILLION REASONS • FOOD PANTRY

Queens, NY

Co-President, Volunteer

9/2022 – 6/2025

- Founded and led citywide 40+ teen initiative distributing food and clothes at NYC's largest food pantry every Saturday.

AWARDS AND INTERESTS

Awards: Wingate Memorial Award (2025), National Latin Exam Silver Medal (2024, 2025), AP Scholar with Honor (2024)

Interests: *New York Times* Games, Sports (Football, Soccer, Hockey, Skiing), Korean Culture, Cooking, Mahjong